

# CS215 Assignment-3

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## 1. Finding optimal bandwidth

### Task 1

#### Part (a)

#### Solution:

We are asked to prove the following:

$$\int \hat{f}(x)^2 dx = \frac{1}{n^2 h} \sum_{j=1}^m v_j^2$$

where  $\hat{f}(x)$  is the histogram estimator,  $v_j$  is the number of points in the  $j$ -th bin, and  $h$  is the bin width.

#### Proof

The histogram estimator  $\hat{f}(x)$  is defined as:

$$\hat{f}(x) = \sum_{j=1}^m \frac{p_j}{h} \mathbb{I}(x \in B_j)$$

also,

$$p_j = \frac{v_j}{n}$$

so the expression becomes,

$$\hat{f}(x) = \sum_{j=1}^m \frac{v_j}{nh} \mathbb{I}(x \in B_j)$$

where: -  $v_j$  is the number of data points in the  $j$ -th bin. -  $\mathbb{I}(x \in B_j)$  is the indicator function that equals 1 if  $x$  belongs to bin  $B_j$  and 0 otherwise.

Next, we need to compute  $\hat{f}(x)^2$ :

$$\hat{f}(x)^2 = \left( \sum_{j=1}^m \frac{v_j}{nh} \mathbb{I}(x \in B_j) \right)^2$$

Since the indicator functions  $\mathbb{I}(x \in B_j)$  are mutually exclusive (a point  $x$  can only belong to one bin at a time), we can ignore cross-terms in the expansion of the square, leading to,

$$\hat{f}(x)^2 = \sum_{j=1}^m \left( \frac{v_j}{nh} \right)^2 \mathbb{I}(x \in B_j)$$

Now, we integrate  $\hat{f}(x)^2$  over the entire domain of  $x$ :

$$\int \hat{f}(x)^2 dx = \int \sum_{j=1}^m \left( \frac{v_j}{nh} \right)^2 \mathbb{I}(x \in B_j) dx$$

Since the bins are disjoint, the integral over each bin  $B_j$  simplifies to the width of the bin, which is  $h$ . Therefore, the integral becomes:

$$\int \hat{f}(x)^2 dx = \sum_{j=1}^m \left( \frac{v_j}{nh} \right)^2 h$$

Simplifying further:

$$\int \hat{f}(x)^2 dx = \frac{1}{n^2 h} \sum_{j=1}^m v_j^2$$

Thus, we have proven that:

$$\int \hat{f}(x)^2 dx = \frac{1}{n^2 h} \sum_{j=1}^m v_j^2$$

## Part (b)

### Solution:

We need to prove the following expression:

$$\sum_{i=1}^n \hat{f}_{(-i)}(X_i) = \frac{1}{(n-1)h} \sum_{j=1}^m (v_j^2 - v_j)$$

where  $\hat{f}_{(-i)}$  is the histogram estimator after removing the  $i$ -th observation, and  $v_j$  is the number of points in the  $j$ -th bin.

## Proof

The histogram estimator  $\hat{f}(x)$  for the complete dataset is given by:

$$\hat{f}(x) = \sum_{j=1}^m \frac{v_j}{nh} \mathbb{I}(x \in B_j)$$

where  $v_j$  is the number of points in the  $j$ -th bin.

When we remove the  $i$ -th observation, the number of points in the bin containing  $X_i$  is reduced by 1. Therefore, the histogram estimator after removing the  $i$ -th point which maybe in any of the bin as indicated by introduction of predicate for the same,  $\hat{f}_{(-i)}(x)$ , is:

$$\hat{f}_{(-i)}(x) = \sum_{j=1}^m \frac{v_j - \mathbb{I}(X_i \in B_j)}{(n-1)h} \mathbb{I}(x \in B_j)$$

Now, sum the values of  $\hat{f}_{(-i)}(X_i)$  for all  $i$ :

$$\sum_{i=1}^n \hat{f}_{(-i)}(X_i) = \sum_{i=1}^n \sum_{j=1}^m \frac{v_j - \mathbb{I}(X_i \in B_j)}{(n-1)h} \mathbb{I}(X_i \in B_j)$$

For each bin  $j$ , the indicator function  $\mathbb{I}(X_i \in B_j)$  equals 1 if  $X_i$  is in bin  $j$ , and 0 otherwise. Therefore, for each bin  $j$ , the sum over  $i$  simplifies to:

$$\sum_{i=1}^n \hat{f}_{(-i)}(X_i) = \frac{1}{(n-1)h} \sum_{j=1}^m \sum_{i \in B_j} (v_j - 1)$$

For each bin  $j$ , there are  $v_j$  points in total. Thus, the sum over  $i \in B_j$  yields  $v_j$  terms, each equal to  $v_j - 1$ . Hence, the inner sum simplifies to:

$$\sum_{i \in B_j} (v_j - 1) = v_j(v_j - 1)$$

Substituting this into the previous expression, we get:

$$\sum_{i=1}^n \hat{f}_{(-i)}(X_i) = \frac{1}{(n-1)h} \sum_{j=1}^m v_j(v_j - 1)$$

Expanding the product  $v_j(v_j - 1)$ , we get:

$$v_j(v_j - 1) = v_j^2 - v_j$$

Thus, the final expression becomes:

$$\sum_{i=1}^n \hat{f}_{(-i)}(X_i) = \frac{1}{(n-1)h} \sum_{j=1}^m (v_j^2 - v_j)$$

## Task 2

### Part(a)

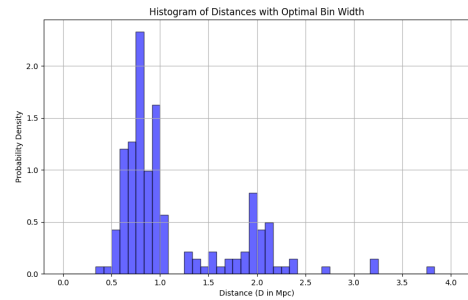
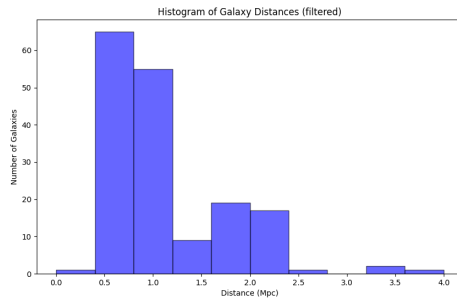
#### Solution:

The probability distributions  $p_j$  for all the bins as calculated in the code as  $\frac{v_j}{n}$  are as follows:

| Bin Number | Bin Range   | Probability |
|------------|-------------|-------------|
| Bin 1      | 0.00 - 0.40 | 0.0059      |
| Bin 2      | 0.40 - 0.80 | 0.3824      |
| Bin 3      | 0.80 - 1.20 | 0.3235      |
| Bin 4      | 1.20 - 1.60 | 0.0529      |
| Bin 5      | 1.60 - 2.00 | 0.1118      |
| Bin 6      | 2.00 - 2.40 | 0.1000      |
| Bin 7      | 2.40 - 2.80 | 0.0059      |
| Bin 8      | 2.80 - 3.20 | 0.0000      |
| Bin 9      | 3.20 - 3.60 | 0.0118      |
| Bin 10     | 3.60 - 4.00 | 0.0059      |

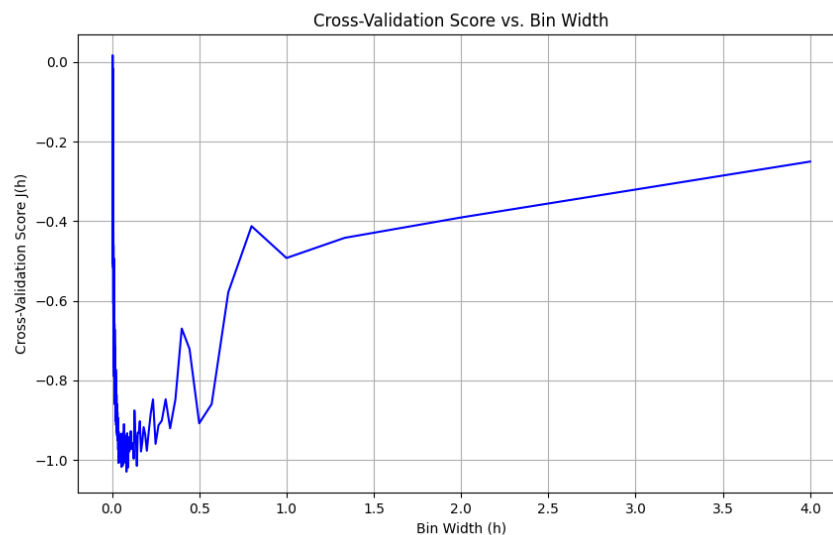
### Part(b)

**Solution:** Bandwidth of the histogram is pretty large and number of bins is also less. Hence the histogram is **oversmooth**.



### Part(d)

**Solution:** Optimal bandwidth is obtained where cross validation score is minimum. From the curve, this is achieved when bandwidth is **around 0.08**. The exact optimal bandwidth from the code come out to be 0.083333 with 48 as number of optimal number of bins.

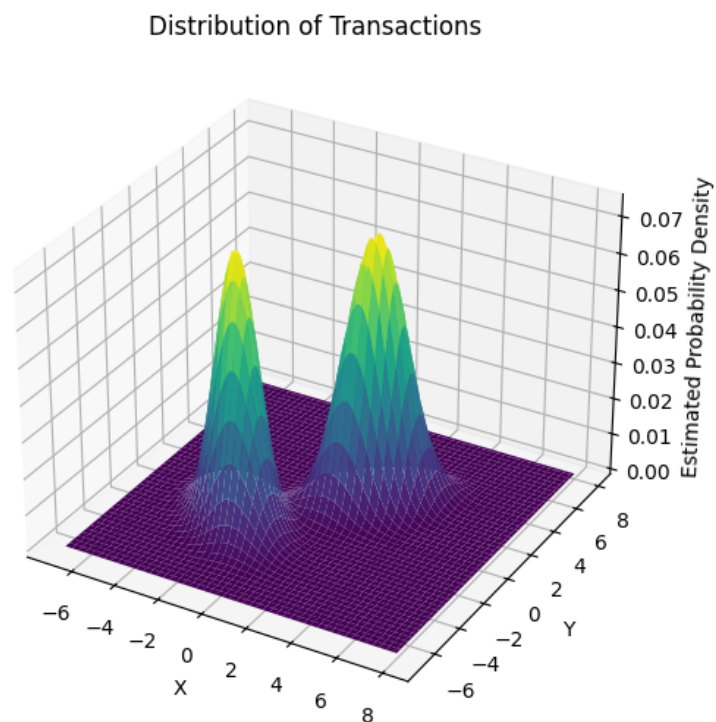
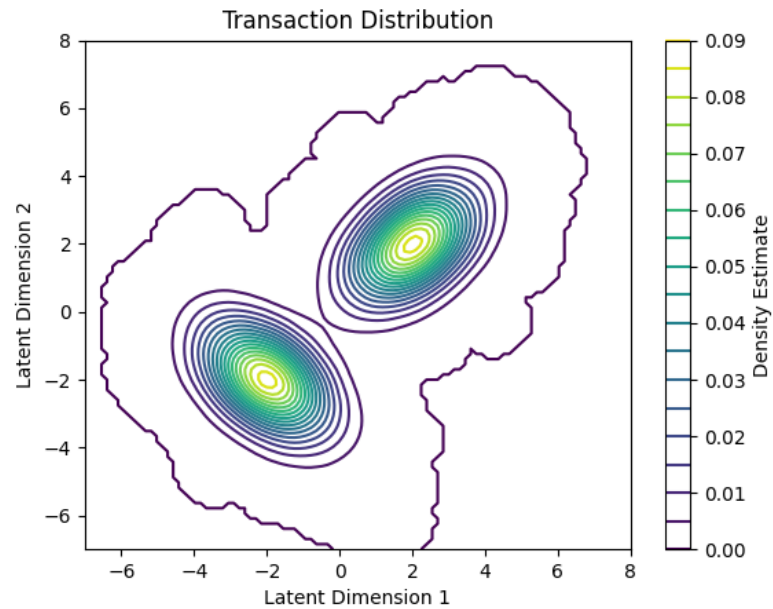


### Part(e)

**Solution:** The major difference between the two is the bandwidth. The plot obtained using optimal number of bins ie 48 provides more finer details about the distributions as using minimised SSR, optimal bandwidth provide correct fit for the histogram. Whereas the one with 10 bins have quite large bandwidth.

## 2. Detecting Anomalous Transactions using KDE

Solution:



From the plots, **the number of modes are 2.**

### 3. Higher Order Regression

**Part(a)**

**Solution:** In a simple linear regression model, the equation of the least squares regression line is:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

where symbols have their usual meanings.

The least squares estimates for  $\hat{\beta}_1$  and  $\hat{\beta}_0$  are given by:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - \bar{x} \sum_{i=1}^n y_i}{\sum_{i=1}^n x_i^2 - n \bar{x}^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where  $\bar{x}$  and  $\bar{y}$  are the means of  $x$  and  $y$ , respectively.

We want to show that the point  $(\bar{x}, \bar{y})$  lies on the regression line. This means that when  $x = \bar{x}$ , the predicted value  $\hat{y}$  should be equal to  $\bar{y}$ .

Substitute  $x = \bar{x}$  into the equation of the regression line:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$$

Now, substitute the expression for  $\hat{\beta}_0$ :

$$\hat{y} = (\bar{y} - \hat{\beta}_1 \bar{x}) + \hat{\beta}_1 \bar{x}$$

Simplifying this:

$$\begin{aligned}\hat{y} &= \bar{y} - \hat{\beta}_1 \bar{x} + \hat{\beta}_1 \bar{x} \\ \hat{y} &= \bar{y}\end{aligned}$$

Thus, when  $x = \bar{x}$ , the predicted value  $\hat{y}$  equals  $\bar{y}$ .

Hence, the point  $(\bar{x}, \bar{y})$  always lies on the least squares regression line, as the predicted value  $\hat{y}$  equals the mean value  $\bar{y}$  when  $x = \bar{x}$ .

### **Part(b)**

**Solution:** The least squares estimates for  $\beta_0$  and  $\beta_1$  can be derived by minimizing the sum of squared residuals:

$$\text{Residual sum of squares (RSS)} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n (Y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2$$

The least squares estimates are:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i Y_i - \bar{x} \sum_{i=1}^n Y_i}{\sum_{i=1}^n x_i^2 - n \sum_{i=1}^n \bar{x}^2}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$$

where  $\bar{x}$  and  $\bar{Y}$  are the means of  $x$  and  $Y$ , respectively.

Now, suppose we use  $z = x - \bar{x}$  as the regressor variable. The new model becomes:

$$Y = \beta_0^* + \beta_1^* z + \epsilon$$

Since  $z = x - \bar{x}$ , this new model can be rewritten as:

$$Y = \beta_0^* + \beta_1^* (x - \bar{x}) + \epsilon$$

Expanding this:

$$Y = \beta_0^* + \beta_1^* x - \beta_1^* \bar{x} + \epsilon$$

We can compare this to the original model  $Y = \beta_0 + \beta_1 x + \epsilon$ . Clearly, the coefficient  $\beta_1^*$  is the same as  $\beta_1$ , i.e.,

$$\beta_1^* = \beta_1$$

Now, to match the original model, we can identify:

$$\beta_0 = \beta_0^* - \beta_1 \bar{x}$$

Thus, the intercept in the transformed model,  $\beta_0^*$ , relates to the intercept in the original model,  $\beta_0$ , as:

$$\beta_0^* = \beta_0 + \beta_1 \bar{x}$$

**Hence the following inferences can be drawn regarding relations between the least square estimates of the two models:**

- The slope  $\beta_1^*$  is the same as  $\beta_1$ :  $\beta_1^* = \beta_1$ .
- The intercept  $\beta_0^*$  is shifted by  $\beta_1 \bar{x}$ :  $\beta_0^* = \beta_0 + \beta_1 \bar{x}$ .

**Differences between the two models are as follows:**

- The new model, by centering the  $x$  values (i.e., subtracting  $\bar{x}$  from each  $x_i$ ), makes the interpretation of the intercept  $\beta_0^*$  simpler as  $\beta_0^*$  represents the value of  $Y$  when  $x = \bar{x}$ , i.e., when  $z = 0$ .
- In the original model,  $\beta_0$  represents the expected value of  $Y$  when  $x = 0$ , which might be less meaningful depending on the context (if 0 is outside the range of observed  $x$  values).

Thus, the transformation to  $z = x - \bar{x}$  simplifies the interpretation of the intercept but leaves the slope unchanged.

## Part (c)

**Solution:** Based on k-folds for better exploration of trained and testing data, optimal degree of best fit comes out to be **5**. Degrees in range  $[4,5,6]$  are more or less best fit for the regression.

Using optimal degree of 5 and plotting the underfit (using degree 2) and overfit (using degree 20), the following are observed:

### Underfit

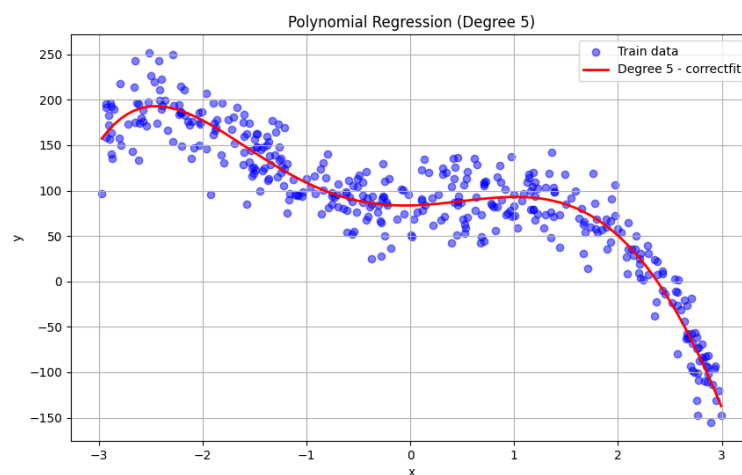
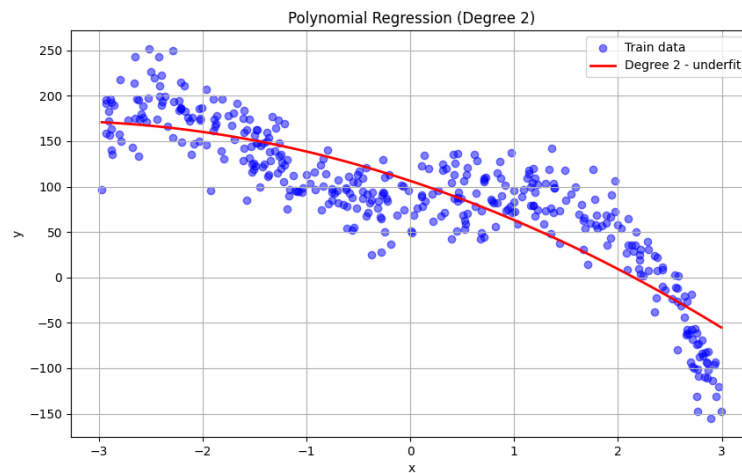
calculated SSR is: 580126, and the calculated  $R^2$  is: 0.6982

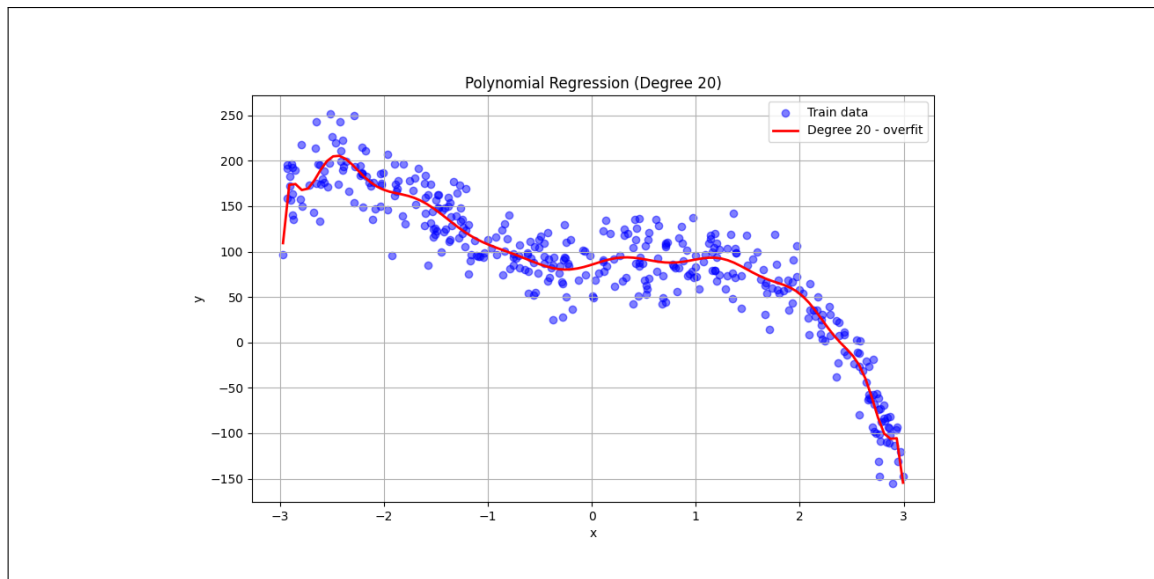
### Correctfit

calculated SSR is: 235760, and the calculated  $R^2$  is: 0.8960

### Overfit

calculated SSR is: 222403, and the calculated  $R^2$  is: 0.9025



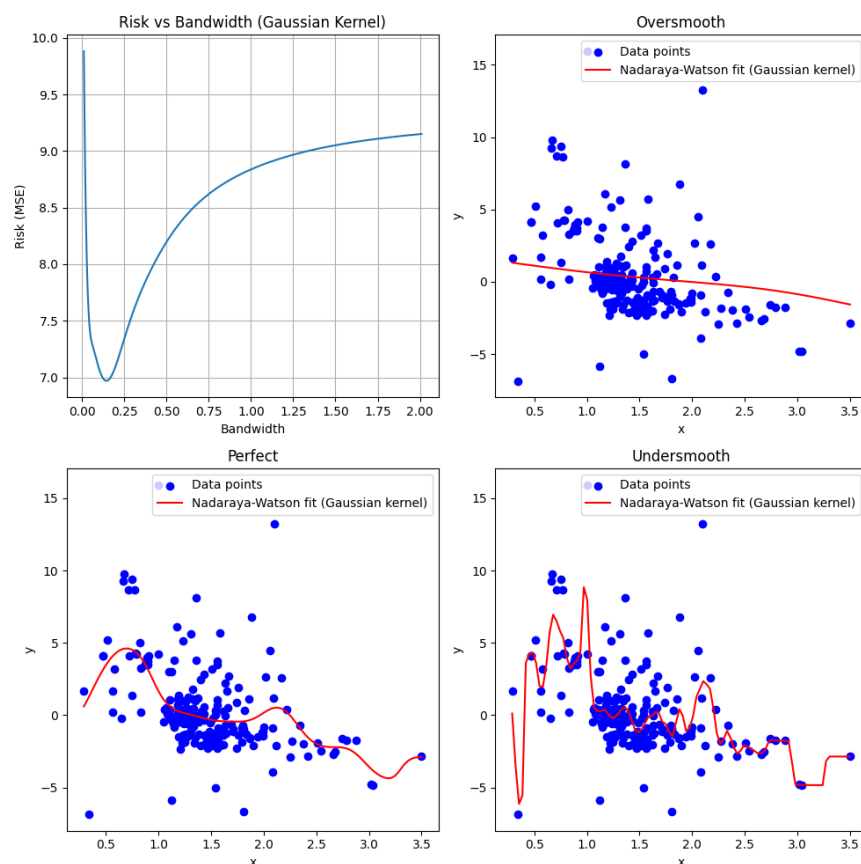


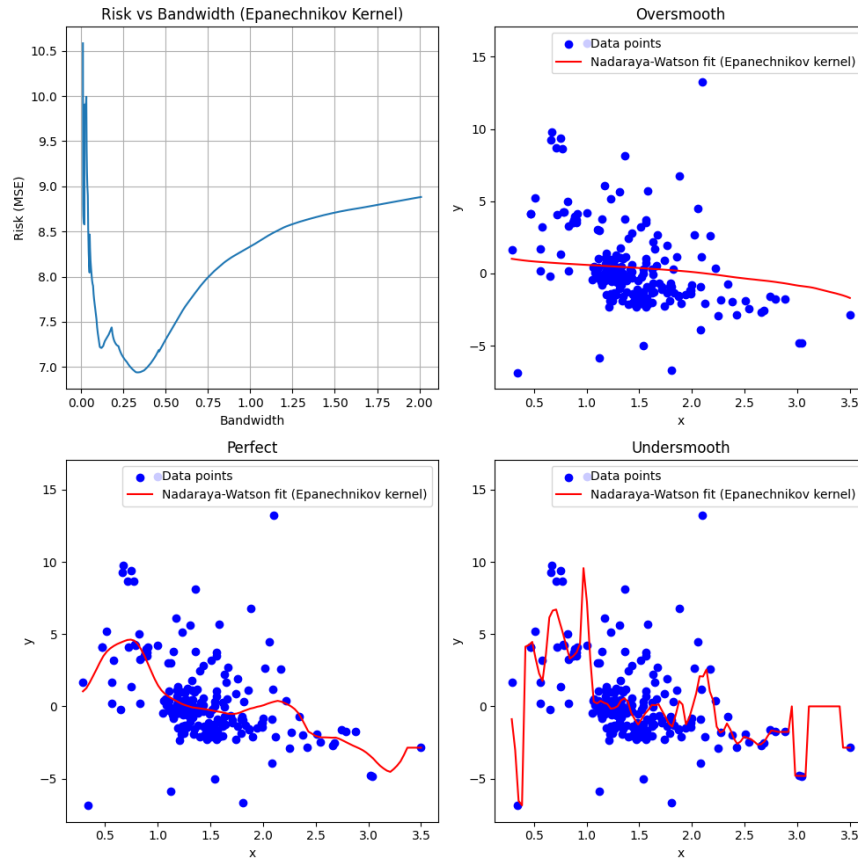
#### 4. Non-parametric regression

##### Task 4.4

##### Part 2(a)

**Solution:**





The provided plots demonstrate why the choice of bandwidth matters more than the choice of kernel:

1. **Risk vs Bandwidth Plot:** The risk is high when the bandwidth is too small or too large, indicating underfitting or overfitting for both the gaussian kernel and Epanechnikov kernel. The minimum risk is observed at a particular bandwidth for both kernels, and we choose which one we get an optimal curve shown in the bottom left, proving bandwidth significantly impacts performance.
2. **Oversmooth:** When choosing a large bandwidth (oversmooth), both fitting curves of the kernels used fail to capture the details in the data, producing flat predictions.
3. **Perfect Fit:** With the optimal bandwidth, both the gaussian and Epanechnikov kernels capture the underlying data structure better than other fits. This causes curves to be over-smooth or under-smoothed.
4. **Under smooth:** When the bandwidth is too small (under smooth), both kernels capture too much noise in the data.

We can conclude that the effect of the kernel is limited compared to the bandwidth itself because bandwidth decides the smoothness of curves. Depending on the

bandwidth, it can make the curve undersmooth, making it susceptible to noise, or oversmooth while being less sensitive but causing a higher bias.

## Part 2(b)

**Solution:** Bandwidth corresponding to minimum estimated risk are as follows:  
**Gaussian Kernel Regression** 0.14406703351675837  
**Epanechnikov Kernel Regression** 0.33016008004002

## Part 3

**Solution:** The famous choices for kernel selection for the Nadaraya-Watson estimator are Gaussian, Uniform, and Epanechnikov.

Here, the sample data is small in size, so significant variations will occur depending on the random shuffle generated by the numpy library. This, in turn, leads to the visible performance difference in the choice of kernels.

Due to the parabolic nature of the **Epanechnikov kernel**, it is more sensitive to local variations, suppressing the noise from points farther away. This makes it efficient in capturing the underlying data structure while reducing the influence of outliers or distant points.

The **Gaussian kernel**, with its bell-shaped curve, assigns higher weights to points near the target and gradually decreases the weight as the distance increases. This makes it sensitive to local variations but with a smooth decay, ensuring that distant points still contribute, albeit minimally. However, this smoothness can lead to better bias-variance trade-offs than the uniform kernel but can still struggle with noise if the data is highly variable.